META GLOBAL AI DEVELOPER HACKATHON · INDIE TRACK

RAN Sentinel

An AI co-engineer that finds a degrading mobile cell, proves the fix is safe and applies it only with an engineer's approval.

Technical documentation · edition of 31 October 2026 · AKMaasif (QuanTell) · MIT licence
github.com/AKMaasif/Meta-Ai-Global-Hackathon-Ran-Sentinel-

This edition opens with an industry chapter (Part A). It explains which problems in mobile network operations RAN Sentinel addresses, how large the market for this kind of software is, what an operator could gain, and how the system would move from a prototype to production. The original technical manual follows unchanged as Part B.

How to read the evidence in Part A

Every figure carries one of four labels, so a reader can tell a published number from a project result or a planning assumption.

SOURCED published by a third party, cited

MEASURED result produced in this project

ASSUMPTION input to a model, to be replaced by pilot data

PROPOSAL not built yet

Contents

Part A · Industry impact and commercial case

A1 Executive summary 2

A2 What the system does today 3

A3 Current challenges in the global telecommunications industry 5

A4 Our proposed solution: transforming telecom operations 8

A5 Global market size, industry value and commercial opportunity 11

A6 Industry transformation potential and competitive advantage 14

A7 Enhancing customer experience 15

A8 Economic benefits and return on investment 17

A9 Competitive analysis and positioning 19

A10 Real-world deployment and scalability 20

A11 Future development roadmap 21

A12 References 22

Part B · Technical manual (edition of 28 Oct 2026) 23

Overview, architecture, agents and models, detection, safety, twin, Open M-Plane, FCC check, customer app, setup, tests and limits

Everything in RAN Sentinel runs on a simulated network. Nothing connects to a real operator, and no endorsement by Meta, any operator or any equipment vendor is implied. Market figures are third-party estimates; financial figures in A8 are illustrative.

PART A · SECTION A1

Executive summary

Mobile networks are run by small teams watching very large numbers of cells. When one cell degrades, an engineer still gathers the evidence by hand, decides the cause and pushes a configuration change, and a wrong change can take down far more than the cell it was meant to fix. RAN Sentinel automates the evidence gathering and the first diagnosis, checks every proposed change with fixed rules and an RF digital twin, and leaves the decision with the engineer.

$1.08T
mobile operator revenues in 2024 SOURCED [1]

15–30%
network opex reduction operators report from AI in operations SOURCED [5]

48 / 48
simulated faults detected, 0 false alarms in 108 cell-days MEASURED

11 / 11
incidents diagnosed correctly by live Muse Spark agents, none wrongly verified MEASURED

The problem

Fault handling in a radio network is still largely manual. Industry surveys put the cost of outages at one to five percent of operator revenue[9], and two of the largest recent outages, at AT&T in 2024 and Optus in 2023, began with a network change rather than a failure[7,8]. Most operators sit between levels 2 and 3 of TM Forum's six-level autonomy scale[6], so the people who decide are also the people who collect the data.

What RAN Sentinel does

It detects a degrading cell with three complementary detectors, has Meta's Muse Spark agents investigate it through read-only tools, and passes the diagnosis through a verifier the agents cannot skip. A plan then goes through nine fixed safety rules and a what-if on Meta's open-source Maveric RF twin before the department that owns it sees a recommendation. An approved change is sent as an Open M-Plane (NETCONF) change package with confirmed-commit, so the radio rolls back by itself if recovery is not confirmed. Customers receive confirmed, plain-language updates; engineers get an FCC Part 4 outage check and an append-only ledger of every AI step.

The opportunity

The closest priced market, software for self-organising and automated radio networks, was about **$7.0 billion in 2025**[13]. We treat the vendor-neutral segment for mid-size and regional operators as the serviceable market (an assumption of 20%, about $1.4 billion) and target 0.5 to 1% of it within five years, roughly $7 to $14 million in annual revenue (A5). For a carrier with 13,500 managed cells, the base-case model returns its first-year cost in about four months, and the conservative case does not pay back, which is exactly what a pilot has to settle (A8).

> **Positioning in one sentence.** RAN Sentinel is not a replacement for a vendor's SON or an operator's OSS; it is a vendor-neutral co-engineer for teams that cannot build their own AI operations stack, with a safety layer that is code, not a model.

PART A · SECTION A2

What the system does today

Before any claim about impact, this section separates what is built and tested from what has only been demonstrated on simulated data and what is proposed. The rest of Part A refers back to this table.

Capability	What it does	Status and evidence
Anomaly detection	Threshold rules, an Isolation Forest and a per-cell, per-hour Mahalanobis baseline; two-window confirmation on 15-minute KPI windows.	MEASURED 48/48 faults, 0 false alarms (simulator); F1 0.61 without labels on TelecomTS[24]
AI investigation	Muse Spark 1.3 agents call read-only tools one at a time (KPIs, alarms, configuration, diagnostics), keep a living plan and rank root causes.	MEASURED live eval 11/11 correct at severity 2; offline 11/11 and 12/12
Diagnosis verifier	Hidden playbook coverage, every stated number looked up in its evidence, root-cause rules and a claims check by a second model. Three refusals escalate to a person.	MEASURED wrong-but-verified 0 in live and offline runs
Field evidence	When data cannot separate two causes, the investigation pauses and asks a field engineer. Photos (SAM 3.1 + vision), CSV, PDF, DOCX and text are read by Muse Spark.	BUILT AND TESTED automated tests; live model check script
Safety rules	Nine fixed rules: action allow-list, step and absolute limits on power and tilt, cause fit, evidence cited, read-only checks first, rollback plan, twin impact.	BUILT AND TESTED unit and end-to-end tests
RF digital twin	Maveric RADP (Meta, Linux Foundation) predicts coverage before and after a tilt or power change; the stricter of Maveric and the simulator copy is used.	MEASURED mean absolute error 1.26 dB against the radio model
Change and rollback	Open M-Plane NETCONF change package, dry run, confirmed-commit with automatic rollback; recovery monitoring.	BUILT AND TESTED against a simulated radio unit, not hardware
Routing and approval	The recommendation goes to the one department that owns it (RAN Optimisation, Field Operations, Regulatory Compliance and others); nothing changes without approval.	BUILT AND TESTED
FCC Part 4 check	Tracks outage user-minutes and opens a compliance ticket with the 120-minute, 72-hour and 30-day deadlines of 47 CFR 4.9(e)[20].	BUILT AND TESTED decision support, not legal advice
Customer care	Customer app with confirmed outage updates, recovery estimate, problem reports into the NOC, outage-credit rule, a Muse Spark concierge and a hands-free voice agent.	BUILT AND TESTED one fictional subscriber
Audit	Append-only Evidence-to-Action Safety Ledger of every detection, tool call, plan, rule, decision and change; shift handover by Muse Wonder.	BUILT AND TESTED
Live operator data, CRM, multi-operator tenancy	Connectors to real PM/FM/CM systems, ticketing and CRM; per-operator isolation.	PROPOSAL see A10 and A11

Inputs and outputs

Inputs

- Cell KPIs every 15 minutes: availability, drop rate, handover success, PRB utilisation, throughput, SINR, uplink interference and others
- Alarms (X.733-style event type and probable cause)
- Radio configuration read over M-Plane (power, tilt, neighbours)
- Field evidence: notes, photos and documents
- Customer problem reports from the app

Outputs

- A confirmed incident with ranked root causes and cited evidence
- A safety-checked change plan with predicted impact, or a dispatch, routed to one department
- A NETCONF change package with automatic rollback, after approval
- Customer updates, an FCC outage assessment and a ledger entry for every step

> **Limits that apply to every claim in Part A.** The network, radio units and customers are simulated, and the operator Northstar Connect is fictional. TelecomTS validates the detection method on real 5G testbed data, not on an operator's cell KPIs. No customer-experience or cost metric has been measured with a real operator.

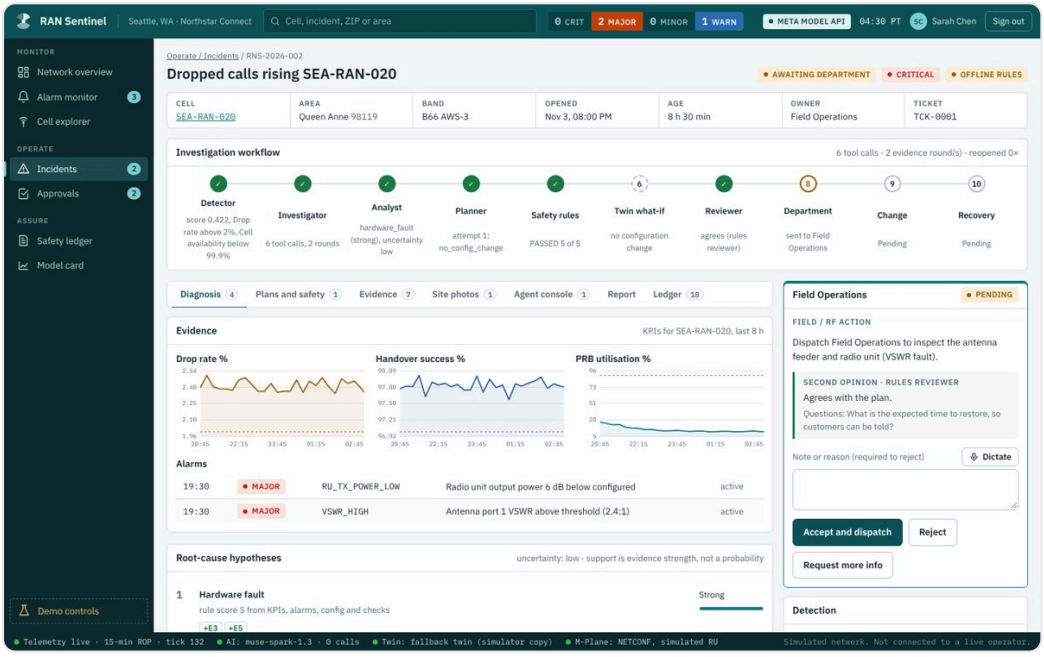


Figure A2.1 The incident workspace: the investigation workflow from detector to recovery, the evidence charts and alarms, and the recommendation waiting for the Field Operations department. Screenshot of the running prototype.

PART A · SECTION A3

Current challenges in the global telecommunications industry

The industry is large and still growing: GSMA counts 5.8 billion unique mobile subscribers and $1.08 trillion of operator revenue in 2024, and mobile technologies contributed $6.5 trillion, or 5.8% of world GDP[1]. Mobile data traffic grew 22% in the year to Q1 2026 and is forecast to reach 515 EB a month by 2031[3], while radio equipment spending is flat, at about $160 billion over 2025–2029[4]. Operators are asked to carry more traffic on networks they are not expanding much, with operations teams that are not growing. The six challenges below are the ones RAN Sentinel can address. Personalisation, customer behaviour analytics and large-scale customer data platforms are outside its scope and are not discussed.

1. Fault triage is manual and slow

What it is	For each degraded cell an engineer opens KPI dashboards, alarm lists and the radio configuration, forms a hypothesis and decides who should act.
Why it occurs	Alarms and KPIs live in different systems; correlating them needs RAN knowledge that is scarce and hard to encode in static rules.
Operators	Engineer hours spent on collection rather than decisions; a backlog when several cells degrade at once.
Customers	Slower data, dropped calls and coverage gaps that last longer than they need to.
Consequences	Operators report 30–70% fewer troubleshooting tickets and 30–40% faster repair once AI-driven operations run at scale[5], which shows how much of today's effort is avoidable.
Why current tools fall short	Threshold alarms catch obvious faults but miss gradual ones; in our simulator, rules alone detected 79% of injected faults (A4).
How RAN Sentinel helps	Three detectors, then Muse Spark agents gather and cite the evidence and rank causes; the engineer reviews a finished case instead of building it.

2. Configuration changes cause some of the worst outages

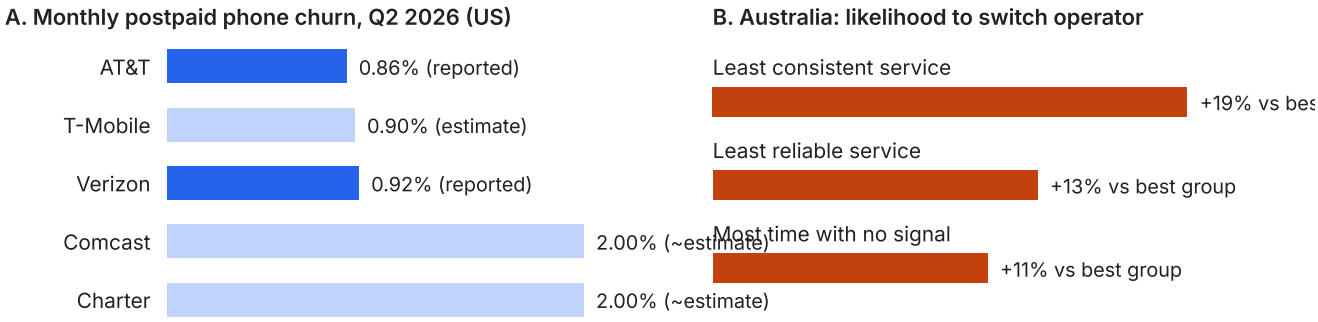
What it is	A change meant to improve one part of the network degrades a much larger part.
Why it occurs	Changes are prepared by hand, peer review is inconsistent, and the effect on neighbours is rarely simulated first.
Real examples	**AT&T, 22 Feb 2024:** an equipment configuration error blocked more than 92 million calls, including 25,000 to 911; rollback took about two hours and full restoration at least twelve. The FCC found peer review missing[7]. **Optus, 8 Nov 2023:** a routine software upgrade triggered routing changes that disconnected about 10.2 million people and 400,000 businesses for up to 14 hours[8].
Operators and customers	Regulatory investigations, compensation, executive departures, and loss of emergency access for customers.
Why current tools fall short	Change processes depend on people remembering to review and test; automation that applies changes without a fixed gate moves the risk rather than removing it.
How RAN Sentinel helps	Every plan passes nine fixed rules and a twin what-if before a person sees it; approval is mandatory; confirmed-commit rolls the radio back automatically if recovery is not confirmed.

3. Outages are expensive and carry reporting duties

What it is	Lost service, repair cost and, in many countries, a legal duty to notify the regulator within hours.
Why it occurs	Congestion, link failures and network failures; a Heavy Reading survey of 54 operators found congestion the most common cause and network failures behind 85% of the most severe outages[9].
Consequences	The same survey estimated outages cost operators more than $20 billion a year, typically 1–5% of revenue. In the US, a wireless outage of 30 minutes affecting 900,000 user-minutes must be notified to the FCC within 120 minutes[20]. The Heavy Reading figure dates from 2016 and is the latest public survey we found.
Why current tools fall short	User-minute counts are usually assembled after the event from several systems, under deadline pressure.
How RAN Sentinel helps	Shorter time from symptom to safe fix, and a live FCC Part 4 check that opens a compliance ticket with its deadlines as soon as the threshold is crossed.

4. Poor network experience drives churn

What it is	Customers leave after repeated bad experiences. US postpaid phone churn was 0.86–0.92% a month at the national carriers in Q2 2026 and about 2% at the cable entrants[10].
Evidence	Opensignal found Australian users with the least consistent service 19% more likely to switch, and those with the least reliable service 13% more likely[11]. ACSI reports that US customers now see network quality and reliability as a baseline expectation; wireless satisfaction reached 77 out of 100 in 2026[12].
Consequences	Each lost subscriber is lost recurring revenue plus acquisition cost to replace them.
How RAN Sentinel helps	Finding and fixing degradations sooner shortens bad-experience periods; whether that measurably lowers churn must be shown in a pilot (A7).



Sources: Recon Analytics in Light Reading (Jul 2026); Opensignal (Apr 2025).

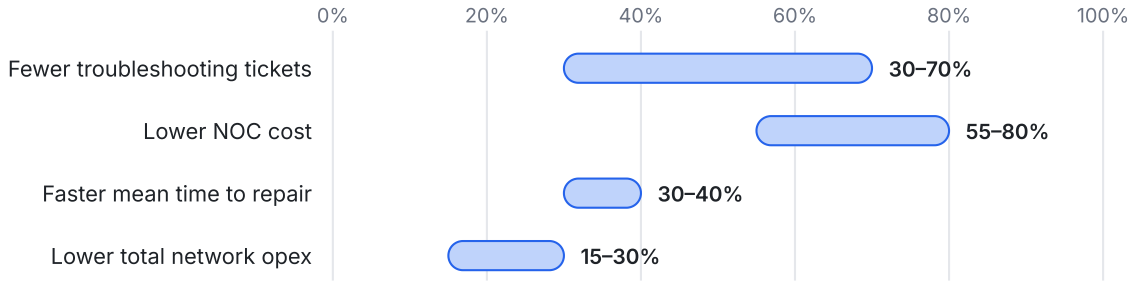
Figure A3.1 Churn levels in the US and the link between network experience and switching in Australia. SOURCED [10,11]

5. Customers learn about outages late and in technical language

What it is	During an incident, customers see a status page that says little, or call a care centre that knows less than the NOC.
Why it occurs	The NOC's incident record and the customer channels are separate systems; publishing a cause before it is confirmed is risky.
Operators and customers	Avoidable care calls, outage credits handled by hand, and customers who feel uninformed.
How RAN Sentinel helps	The same incident drives customer updates, but only after engineers confirm the cause; a filter removes technical identifiers; the customer sees a recovery estimate, can report a problem straight into the NOC and gets the credit rule applied.

6. Smaller operators cannot build AI operations themselves

What it is	Large groups build their own agentic tools (Deutsche Telekom's RAN Guardian with Google Cloud[18]) or buy them as part of a vendor's management stack[17,19]. Regional, rural and emerging-market operators, from rural US carriers serving fewer than 5,000 customers[25] to mid-size operators in multi-vendor networks, have neither the data-science teams nor a single-vendor network.
Evidence	TM Forum found 21% of operators at autonomy level 3 or above in 2026; an Accenture study put 79% at level 0 or 1, with 22% expecting level 4 by 2030[6].
Why current tools fall short	Vendor tools are tied to that vendor's SMO or RIC; in-house tools are not for sale.
How RAN Sentinel helps	Vendor-neutral by design: it reads KPIs and alarms, talks Open M-Plane to the radio, runs on a laptop without a GPU and needs no labelled fault history to start.



Ranges operators reported to McKinsey (2026) for AI-driven network operations at scale. Not RA

Figure A3.2 What operators report after deploying AI-driven network operations at scale. These are industry figures, shown to size the problem; they are not RAN Sentinel results. SOURCED [5]

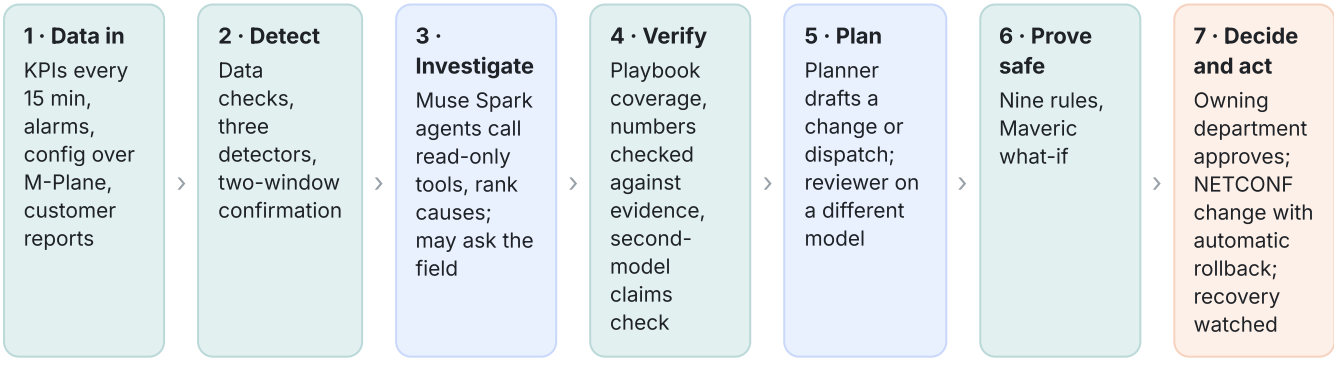
PART A · SECTION A4

Our proposed solution: transforming telecom operations

RAN Sentinel keeps the operator's existing systems and people and changes who does the gathering, checking and paperwork. The table maps each challenge from A3 to the capability that addresses it and to the benefit we expect, with the evidence behind it.

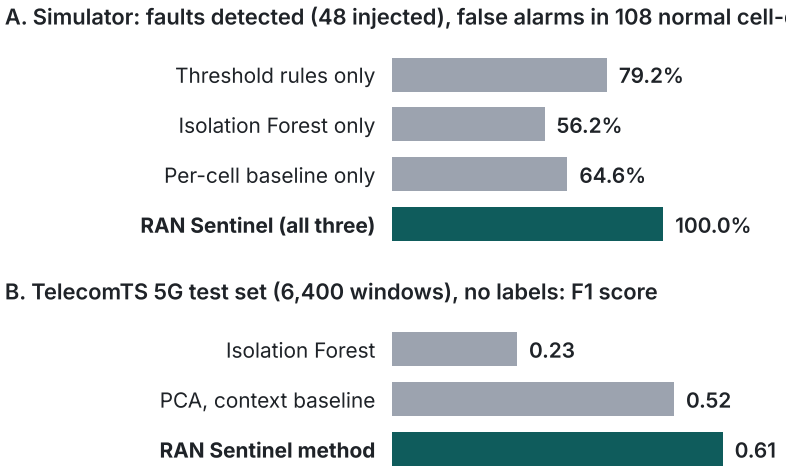
Industry problem	Existing approach	RAN Sentinel	Expected benefit
Manual triage	Engineer correlates dashboards, alarms and config per cell	Three detectors; agents gather cited evidence through read-only tools; verifier checks every number	Engineer time moves from collection to decisions. MEASURED 11/11 correct live; ASSUMPTION 50% time saved (A8)
Missed gradual faults	Static thresholds	Rules + Isolation Forest + per-cell, per-hour baseline with two-window confirmation	MEASURED 100% vs 79% detection, 0 false alarms (simulator)
Risky changes	Manual method of procedure, peer review	Nine fixed rules, Maveric what-if, mandatory approval, confirmed-commit rollback	Unsafe plans never reach a person; failed changes revert. BUILT AND TESTED
Wrong team, slow hand-off	Tickets bounced between teams	One department per recommendation, with the evidence attached	Fewer hand-offs. TO BE MEASURED
Outage reporting	User-minutes compiled after the event	Live FCC Part 4 user-minute tracking and deadlines	Earlier, auditable notifications. BUILT AND TESTED
Customer communication	Generic status pages, care calls	Confirmed plain-language updates, recovery estimate, reports into the NOC, voice agent	Fewer "is it just me" calls. TO BE MEASURED
No AI team	Vendor-tied SON or nothing	Vendor-neutral, open interfaces, laptop-scale models, no labels needed	Automation within reach of smaller operators. MARKET ASSUMPTION

How a case moves through the system



Blue: Meta models suggest. Teal: fixed code decides. Orange: a person approves. A model can never apply a change, approve one or skip a rule.

Operators use the results in three places: the **work queue** (a finished case per incident with its evidence and a recommendation), the **approval** step (approve, reject with a reason, or send back for more information) and the **ledger**, which an auditor can read later. Customer channels draw on the same incident once its cause is confirmed.



Why it matters
Threshold rules are the usual first line in a NOC. Alone they missed 10 of 48 faults here.

Combining rules, an Isolation Forest and a per-cell, per-hour baseline caught all 48 with no false alarms.

Measured in this project; the simulator is our own.

Figure A4.1 RAN Sentinel's detector compared with the methods it combines, which are common in operations today. Panel A: our 36-cell simulator, 4 fault types × 3 severities × 4 cells. Panel B: the open TelecomTS 5G dataset[24], detector chosen on a validation split, never on test. MEASURED

Integration with an operator's systems

RAN Sentinel sits beside the existing OSS and talks to it through interfaces operators already have. It needs read access to performance and fault data, read and write access to radio configuration through a standard management plane, and a link to the ticketing and customer channels. Nothing in the radio network has to be replaced.

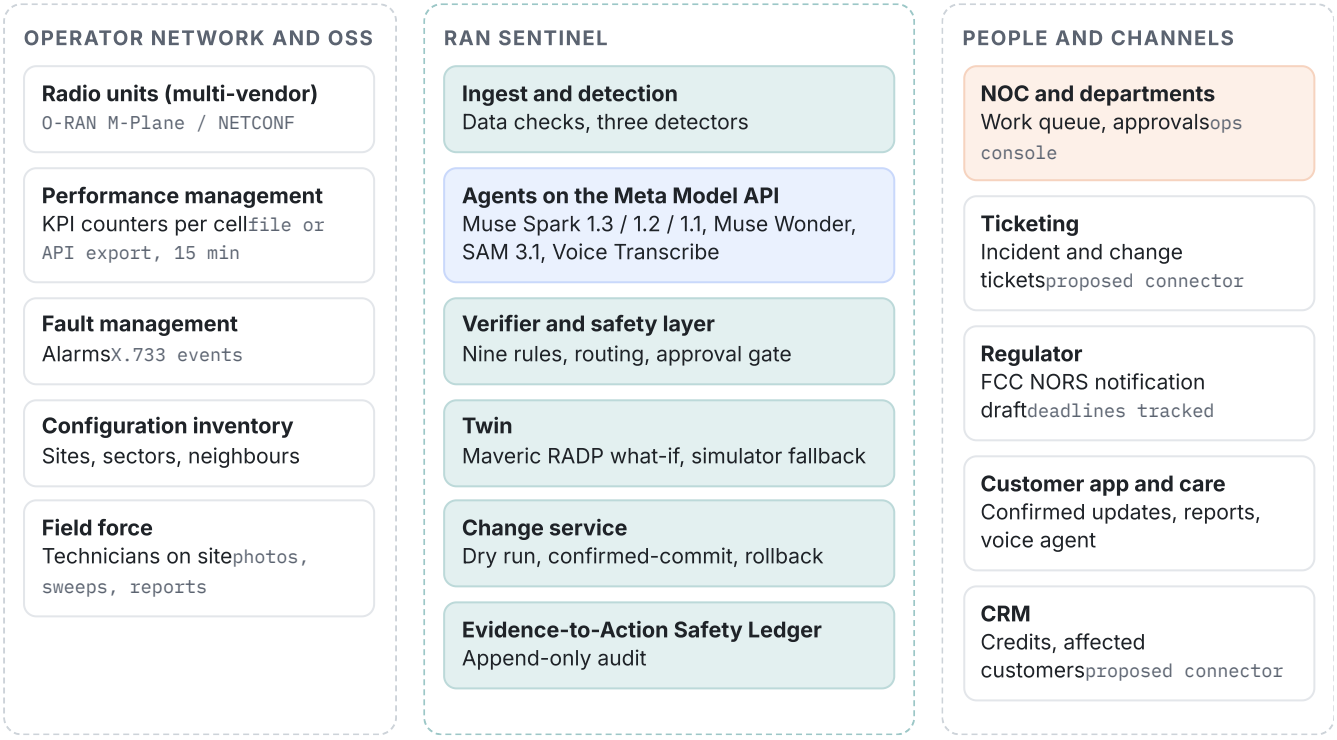


Figure A4.2 Proposed integration with an operator. Interfaces shown in grey are proposals; the prototype uses a simulated network, a simulated radio unit and its own customer app.

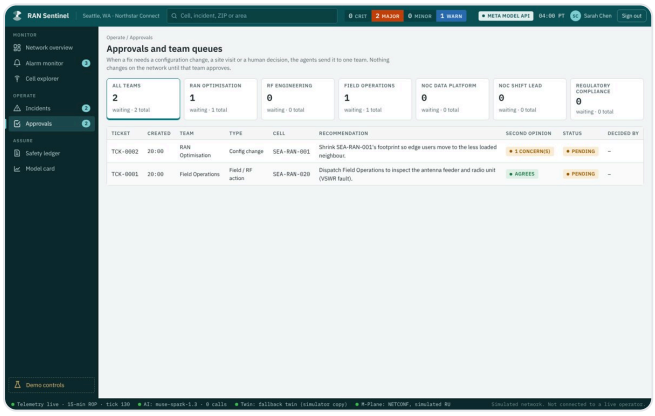


Figure A4.3 Approvals: each recommendation waits for the department that owns it.

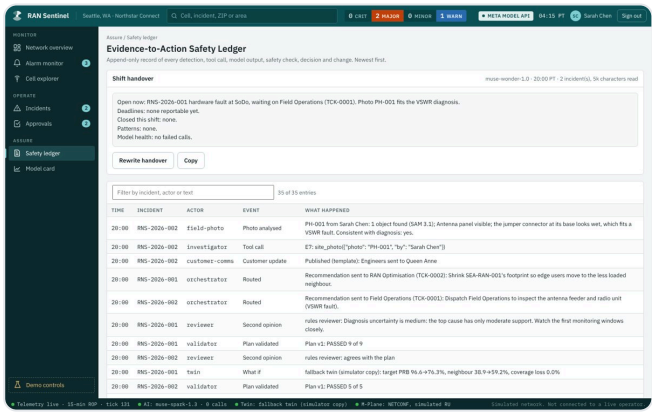


Figure A4.4 The safety ledger: every detection, tool call, rule result, decision and change.

PART A · SECTION A5

Global market size, industry value and commercial opportunity

A. The telecommunications market

Two widely cited measures describe the industry. Grand View Research put global telecom services, fixed and mobile, at $1,805.6 billion in 2022 and $1,885.4 billion in 2023, growing 6.2% a year to $2,874.8 billion in 2030, with Asia Pacific above 33% of the total and growing fastest at 7.0%[2]. GSMA Intelligence, which counts mobile operators only, reports $1.08 trillion of operator revenue in 2024, rising to $1.25 trillion in 2030, and $1.3 trillion of operator capital expenditure over 2024–2030[1]. The two differ because the first includes fixed broadband, enterprise and pay-TV services and the second does not.

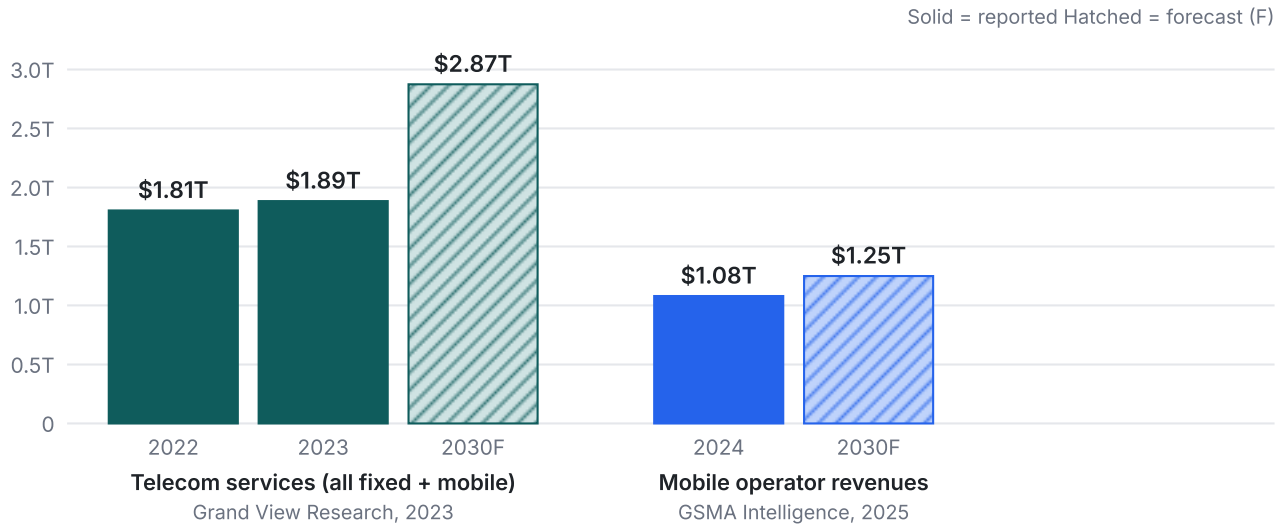


Figure A5.1 Global telecom market, two definitions. Values in USD, worldwide. SOURCED [1,2]

Growth drivers. Data traffic (+22% a year[3]), 5G growing from 25% to 57% of connections by 2030[1], fixed wireless access, and pressure to lower operating cost while radio spending stays flat[4]. The last point matters most here: if operators cannot grow revenue quickly, the remaining lever is the cost of running the network.

B. The market for this kind of solution

RAN Sentinel belongs to **AI-driven radio network operations**: automated detection, diagnosis and optimisation of the RAN. No research firm publishes that exact segment, so we use two neighbouring ones and show how far apart the estimates are.

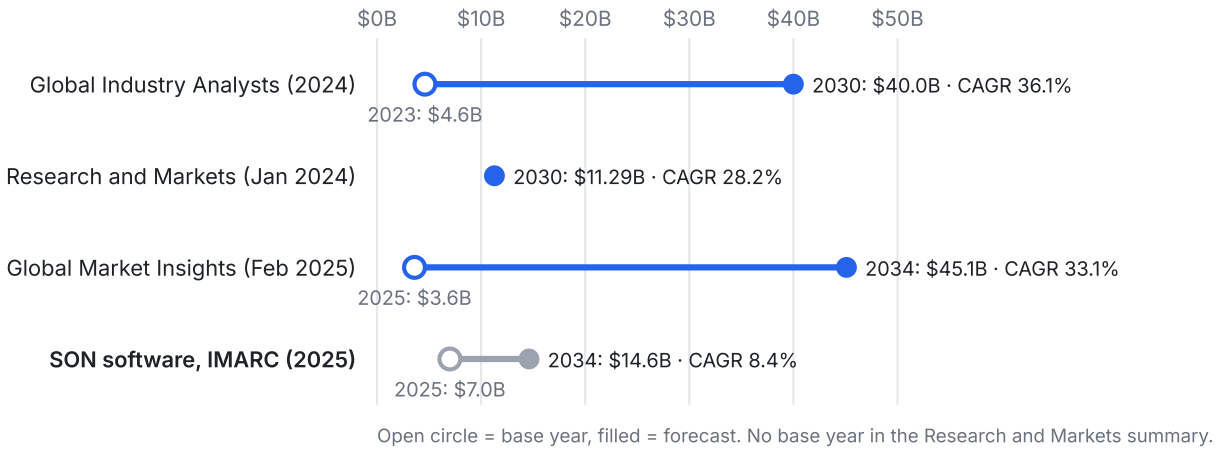


Figure A5.2 Published estimates for AI in telecommunications and for SON software. Values in USD, worldwide. SOURCED [13–16]

The AI-in-telecom estimates range from $3.6 to $4.6 billion today and from $11 to $45 billion at the end of their forecasts. They differ because each firm defines the market differently: some count customer analytics and virtual assistants, some count only software, and their base years differ. The GMI summary is also internally inconsistent about its 2024 value. The SON estimate is narrower but closer to what RAN Sentinel does: software that plans, configures, optimises and heals radio networks. Nokia's MantaRay SON at NTT DOCOMO and Ericsson's agentic rApps are examples of what it covers[17,19]. We size the opportunity on SON, because it is the most conservative relevant figure.

C. TAM, SAM and SOM

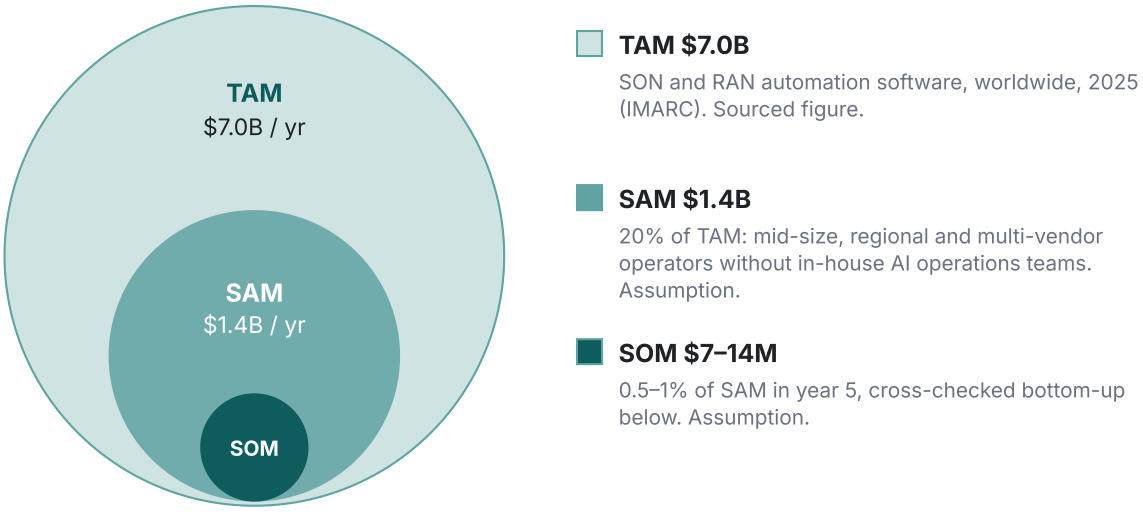


Figure A5.3 Addressable market. TAM is a sourced figure; SAM and SOM are our assumptions, stated below so they can be challenged.

Level	Value	Definition and basis	Type
TAM	$7.0B	Worldwide spending on SON and RAN automation software in 2025 (IMARC[13]), forecast at 8.4% a year to $14.6 billion in 2034. Excludes RAN hardware ($160 billion over 2025–29[4]) and general telecom revenue.	SOURCED

SAM	$1.4B	The share bought by operators outside the largest groups, in multi-vendor networks, without an in-house AI operations team: regional and rural carriers, MVNO-hosting networks and mid-size operators in emerging markets. GSMA represents more than 750 mobile operators[21]; most are not tier-1 groups. We assume they account for 20% of SON spending.	ASSUMPTION
SOM	$7–14M	0.5–1% of SAM in year five. Bottom-up check: 40 operators of the size modelled in A8 (13,500 cells each) at $1.50 per cell per month gives $9.7 million a year.	ASSUMPTION

> **What we do not claim.** We do not compare RAN Sentinel with the $1–3 trillion telecom market; operators spend only a small share of revenue on this kind of software. The relevant number is the $7 billion SON market, of which a vendor-neutral entrant could realistically reach a small fraction.

Adoption trends and demand

- **Autonomy targets:** TM Forum awarded 37 certifications toward level 4 autonomous operations in late 2025, and DOCOMO, Deutsche Telekom and others state level 4 as a goal[6,19].
- **Agentic AI in vendor portfolios:** Ericsson sells agentic rApps as a service on AWS (Feb 2026)[17]; Nokia's MantaRay SON runs closed-loop changes without human intervention[19].
- **Regulation:** outage reporting rules such as 47 CFR 4.9 make an auditable trail of who changed what, and why, a requirement rather than a nice-to-have[20].

PART A · SECTION A6

Industry transformation potential and competitive advantage

The realistic path for RAN Sentinel is not to replace an operator's OSS or a vendor's SON, but to become the co-engineer that sits between monitoring and change for operators who have neither. The transformation it enables is a change in the engineer's job: from assembling evidence to judging a finished case.

What changes in operations

- **Automation of the repetitive part:** detection, evidence gathering, first diagnosis, impact prediction, routing, ticket text, outage counting and customer updates.
- **Data-driven decisions:** every recommendation cites its evidence; numbers the model states are checked against the data before anyone sees them.
- **Proactive care:** customers in an affected area are told once a cause is confirmed, before they call.
- **Faster response:** the case is ready when the engineer opens it; a failed change reverts without waiting for a person.
- **Labels over time:** each incident an engineer confirms in the ledger becomes a training label; on TelecomTS, labels raised F1 from 0.61 to 0.97 MEASURED.

Why it can scale across operators

- **Open interfaces:** Open M-Plane (O-RAN, NETCONF/YANG) for configuration, standard KPI and alarm feeds, and Maveric for the twin, both from Meta's open telecom work.
- **No labelled history needed:** the detectors learn normal behaviour per cell and hour, so an operator can start on day one.
- **Light footprint:** the models run on a laptop without a GPU; the Meta Model API is called only for investigation.
- **Emerging markets:** mid-size operators in multi-vendor LTE networks, often run by small teams, are where a vendor-neutral, low-cost tool fits best. The author is an LTE RAN engineer who works on such networks.

Path to adoption

Area	What it takes
Technical requirements	Read access to PM and FM data, M-Plane access to radios (or a vendor EMS adapter where M-Plane is not available), a Meta Model API account or a hosted model, and a place to run a Python service and SQLite or Postgres.
Integration challenges	Vendor-specific counter names and alarm texts must be mapped; older radios may expose only proprietary management interfaces; ticketing and CRM connectors are per operator.
Regulation and governance	Outage reporting (for example FCC Part 4), data residency for subscriber data, change-management policies, and an audit trail of AI decisions. The ledger and mandatory approval are designed for this.
Partnerships	System integrators that already serve regional operators; associations such as the Competitive Carriers Association in the US; open-source communities around Open M-Plane and Maveric (Linux Foundation).
Differentiation	A safety layer that is fixed code, independent reviewer model, confirmed-commit rollback, vendor neutrality, and one incident record shared by the NOC, compliance and customer care.

What would stop adoption. Operators will not let software near their radios without a long trust-building period. That is why the first deployment phase is read-only and every change needs approval (A10). A vendor that bundles similar capability into the management system an operator already owns is the strongest competitive threat.

PART A · SECTION A7

Enhancing customer experience through our proposed model

Most of RAN Sentinel's effect on customers is indirect: degradations found and fixed sooner, and fewer changes that go wrong. The customer app adds a direct effect: clear, confirmed information during an incident and an easy way to report a problem. None of the customer metrics below has been measured with real customers yet.

Detect early
Gradual faults found by the baseline detector, not by complaints

Diagnose right
Verified cause before any plan; field evidence when data is not enough

Fix safely
Rules, twin and rollback keep a fix from creating a wider outage

Tell clearly
Confirmed updates, recovery estimate, voice agent, no jargon

Close the loop
Customer reports feed the NOC; credit rule applied automatically

Figure A7.1 Customer experience framework: five stages, each tied to a built capability.

Improvement	Customer problem today	Capability	Expected benefit
Fewer, shorter interruptions	Slow data or drops for hours before anyone acts	Three detectors and an investigation that starts automatically	Shorter degradations
Right fix first time	A technician is sent for a fault that was a config issue, or the reverse	Verifier, field-evidence pause, cause-fit safety rule	Fewer repeat incidents
No collateral outages	A change elsewhere breaks service	Nine rules, twin what-if, automatic rollback	Better reliability
Clear communication	Status pages that say nothing; care agents without NOC data	Confirmed plain-language updates and recovery estimate in the app	Less uncertainty, fewer calls
Easy reporting	Long phone queues to report a problem	In-app report and voice agent that send the report to the NOC as evidence	Lower customer effort
Fair compensation	Credits require a call and a dispute	Outage-credit rule (30 minutes or more) applied to confirmed incidents	Trust and transparency

Before and after: one customer, one outage

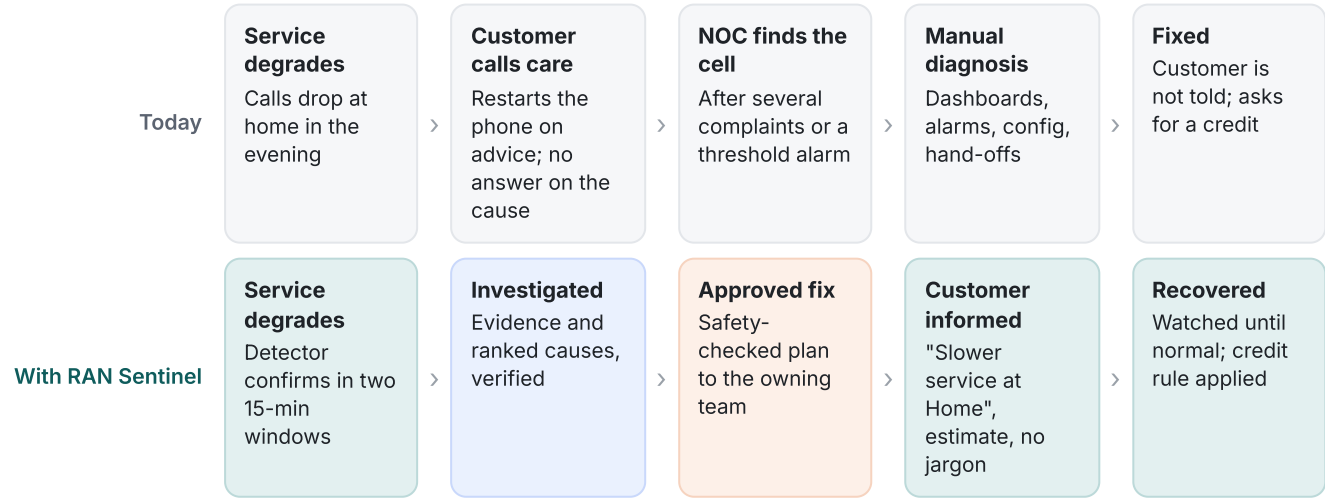


Figure A7.2 Customer journey compared. Step timings depend on the operator and are not claimed.

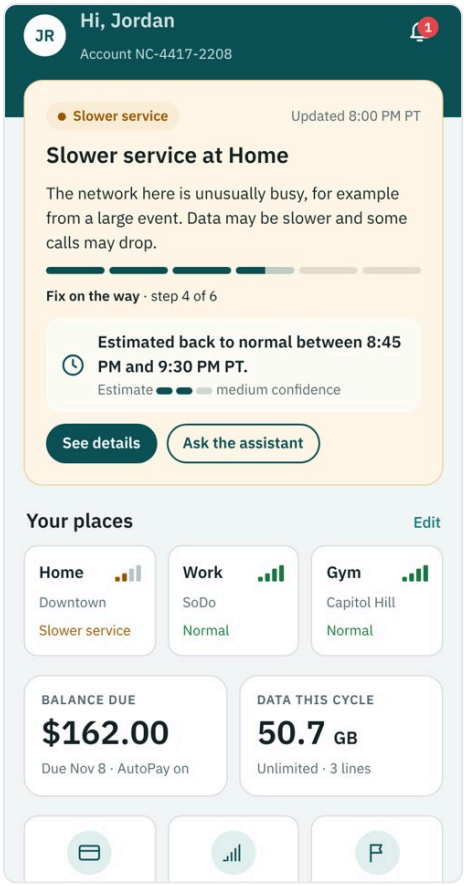


Figure A7.3 Northstar Connect customer app during an incident: a confirmed, plain-language status, a repair tracker and a recovery estimate.

How improvements would be measured

All metrics below need a pilot with a real operator. The project supplies the data each one needs; it does not yet supply results.

Metric	How RAN Sentinel feeds it	Status
Mean time to detect / repair	Ledger timestamps from first abnormal window to recovery	PILOT
Service availability	Cell availability KPI before and after	PILOT
Outage user-minutes	FCC check counts them per incident	PILOT
First contact resolution	Care sees the confirmed incident at first contact	PILOT
Average resolution time	Report-to-recovery time per customer report	PILOT
CSAT, CES	Short survey after an incident closes in the app	PILOT
NPS	Periodic survey, affected vs unaffected areas	PILOT
Churn rate	Churn in areas with incidents, before and after, against a control region	PILOT
Change failure rate	Share of approved changes rolled back	PILOT

A credible pilot compares a treated region with a control region over the same months, because churn and satisfaction move with seasons, prices and handset launches.

PART A · SECTION A8

Economic benefits and return on investment for telecom operators

The model below is **illustrative**. It uses a hypothetical regional carrier and inputs we chose to be plausible; every input is meant to be replaced by numbers from a pilot. It counts two benefits that can be priced (engineer time and retained subscribers) and leaves out others that are real but harder to value: avoided outage costs, regulatory penalties, fewer failed changes and lower care volume.

Input	Conservative	Base	Best	Type
Managed cells (1,500 sites × 3 sectors × 3 carriers)	13,500			ASSUMPTION
Subscription per cell per month	$1.50			ASSUMPTION
Degraded-cell incidents per day	25	40	60	ASSUMPTION
Engineer time per incident today	45 min at $75 per loaded hour			ASSUMPTION
Share of that time saved	30%	50%	60%	ASSUMPTION
Monthly churn reduction (percentage points)	0.00	0.01	0.02	ASSUMPTION
Subscribers; ARPU; contribution margin	600,000; $45 per month; 40%			ASSUMPTION
One-time implementation (integration, pilot)	$150,000	$100,000	$75,000	ASSUMPTION
Hosting; model API per incident	$24,000 per year; $0.50			ASSUMPTION

Formulas

```
Time saving = incidents/day × 365 × 0.75 h × share saved × $75
Retention = subscribers × churn reduction × 12 months × ARPU × 12 months × margin
Running cost = cells × $1.50 × 12 + hosting + incidents/day × 365 × API cost
Payback (months) = implementation ÷ ((benefit − running cost) ÷ 12)
3-year ROI = (3 × benefit − implementation − 3 × running cost) ÷ (implementation + 3 × running cost)
```

Result (per year unless stated)	Conservative	Base	Best
Subscription	$243,000	$243,000	$243,000
Hosting and model API	$28,562	$31,300	$34,950
Running cost	**$271,562**	**$274,300**	**$277,950**
Engineer time saved	$153,984	$410,625	$739,125
Subscribers retained	0	720	1,440
Retention value	$0	$155,520	$311,040
Total benefit	**$153,984**	**$566,145**	**$1,050,165**
Net annual benefit	**−$117,578**	**$291,845**	**$772,215**
One-time implementation	$150,000	$100,000	$75,000
Payback	does not pay back	4.1 months	1.2 months
3-year ROI	**-52%**	**+84%**	**+247%**
Break-even incidents/day (engineer time only)	44	27	23

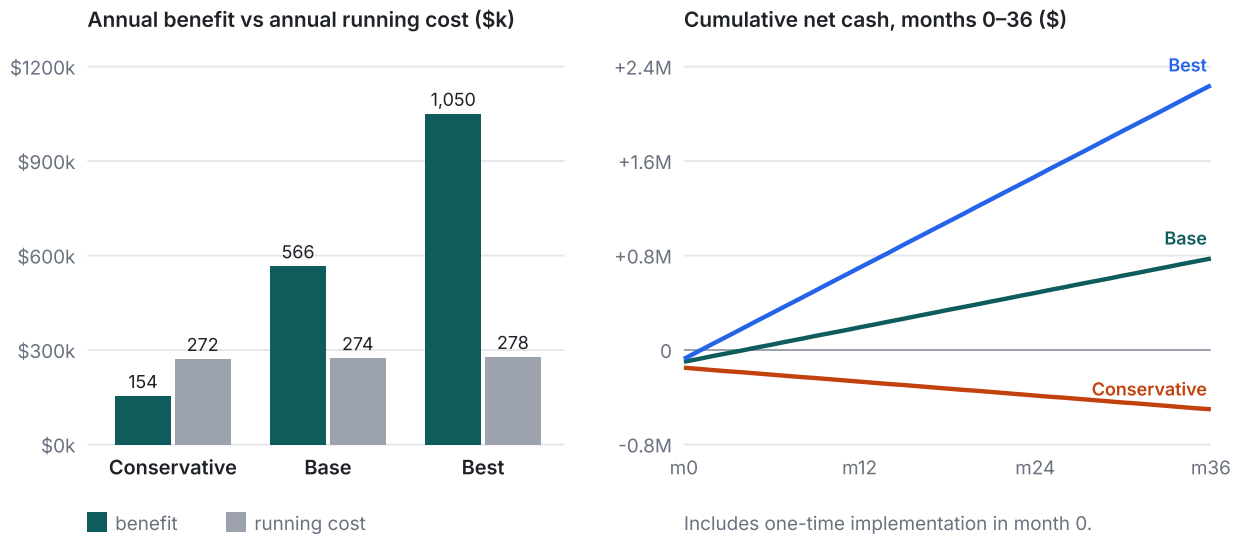


Figure A8.1 Annual benefit against running cost, and cumulative net cash over three years, for the three scenarios. ILLUSTRATIVE

Reading the result

- The **base case** pays back in about 4 months, with a three-year ROI of 84%. About 27% of its benefit comes from the churn assumption, the least certain input.
- The **conservative case does not pay back**: at 25 incidents a day and 30% time saved, engineer time is worth less than the subscription. On engineer time alone, the subscription breaks even at about 27 incidents a day at 50% saved. The price and the incident volume, not the technology, decide whether the product makes sense for a given operator.
- The **best case** pays back within weeks, but depends on high incident volumes and a measurable churn effect.

Other potential value, not priced

- **Avoided outages:** if outages cost 1–5% of revenue[9], avoiding even one change-induced outage a year can outweigh the subscription.
- **Regulatory exposure:** timely, complete outage notifications and an audit trail.
- **Care volume:** fewer calls during incidents when customers are already informed.
- **Resource use:** engineers spend time on planning and fieldwork rather than triage; technicians are dispatched with evidence.

Evidence needed to validate the model

Assumption	How a pilot would measure it
Incidents per day	Count of degraded-cell tickets over three months of the operator's history
Engineer time per incident and share saved	Time-and-motion study on a sample of incidents, with and without RAN Sentinel
Churn reduction	Treated vs control regions over at least two quarters
Implementation cost	Actual integration effort for PM/FM feeds, M-Plane or EMS access, and ticketing
Model API cost	Logged tokens per incident; the prototype already records usage per call

PART A · SECTION A9

Competitive analysis and industry positioning

Agentic and AI-driven RAN operations is an active field, led by the large equipment vendors and the largest operators. RAN Sentinel is not ahead of them on scale, maturity or integration. Its case rests on who it is for and how it proves safety.

	Manual NOC with dashboards	Vendor SON: Nokia MantaRay[19]	Ericsson Agentic rApp as a Service[17]	Deutsche Telekom RAN Guardian[18]	RAN Sentinel
Core capability	Human triage and changes	Planning, provisioning, closed-loop optimisation	Agentic and generative AI rApps for RAN	Agentic detection and remediation for DT's network	Detect, investigate, verify, plan, prove safe, route, change with rollback
Intended users	Any operator	Operators with Nokia SMO; large networks	Operators on Ericsson SMO / Non-RT RIC	Deutsche Telekom (in-house)	Mid-size, regional, multi-vendor operators
Technical approach	Thresholds, experience	AI-driven SON inside the SMO	Agents on AWS, via the RIC	Gemini 2.0 on Google Cloud Vertex AI	Meta models suggest; fixed rules, twin and approval decide
Integration	None	Nokia SMO; multi-vendor stated as future evolution	Ericsson SMO / RIC, AWS Marketplace	Internal	KPI and alarm feeds, Open M-Plane; no RIC or SMO
Deployment complexity	Low	High	Medium to high	Not offered externally	Low (Python service, laptop-scale models)
Strengths	Trusted, flexible	Mature, proven at scale (DOCOMO)	Vendor depth, marketplace delivery	Operator-built, tailored	Safety layer in code, auditable ledger, vendor-neutral, open interfaces
Limitations	Slow, does not scale	Tied to the vendor stack	Tied to the vendor stack	Not for sale	Prototype on simulated data; no live operator, no hardware M-Plane, few connectors

Competitor capabilities are taken from the cited public announcements; we have not tested these products.

Genuine advantages

1. **The safety layer is not AI.** Nine fixed rules, a twin what-if and a dry run sit between the model and the network; three verifier refusals go to a person.
2. **Two models check each other.** The plan reviewer runs on a different Muse Spark version from the planner.
3. **The loop closes.** Confirmed-commit rollback, recovery monitoring, customer updates and the FCC check all come from one incident record.
4. **Built on open pieces.** Open M-Plane and Maveric (Linux Foundation, from Meta) and an open benchmark dataset.

Gaps before commercial deployment

- Validation on a real operator's PM/FM data and on real radios through M-Plane or a vendor EMS.
- Connectors for ticketing, CRM and inventory; per-operator counter and alarm mappings.
- Multi-tenant isolation, single sign-on, roles, HTTPS, hashed credentials, persistent state and high availability.

PART A · SECTION A10

Real-world deployment and scalability

Trust is earned in stages. Each phase widens what RAN Sentinel may do only after the previous phase has produced evidence.

Phase 1 · Prototype validation

0–3 months · read-only

- Replay three months of the operator's PM/FM history
- Measure detection and diagnosis against past tickets
- Map counters and alarms; check data quality
- Shadow mode: recommendations not shown to the NOC

Phase 2 · Pilot

3–9 months · one region

- Live feeds for one region; NOC sees recommendations
- Changes applied by engineers through existing tools
- Treated vs control region for MTTR, churn, CSAT
- Authorised, anonymised data only

Phase 3 · Production

9–18 months · approved changes

- M-Plane or EMS change path with confirmed-commit
- SSO, roles, encryption, secrets management
- Monitoring, logging, high availability, backups
- Ticketing and CRM connectors

Phase 4 · Multi-operator

18 months+

- Tenant isolation, per-operator models and rules
- Vendor adapters beyond Open M-Plane
- Regional regulatory modules beyond FCC Part 4
- Partner-led deployments in new markets

Figure A10.1 Adoption roadmap. PROPOSAL Durations are planning estimates.

Risks and controls

Risk	Description	Control (built or proposed)
Data privacy	Customer reports and locations are personal data; subscriber data may not leave the country	Customer app sees only its own places and published information BUILT; regional hosting, anonymised NOC data, data-processing agreements PROPOSED
Security	Write access to radios is a high-value target	Portal-bound sessions, sign-in lockout, upload hardening, security headers, keys never logged BUILT AND TESTED; SSO, mTLS to the management plane, penetration test PROPOSED
Model error	A wrong but well-formed diagnosis	Verifier, independent reviewer model, nine rules, twin, mandatory approval, rollback BUILT
Prompt injection	Instructions hidden in alarms, tool data or uploaded files	Untrusted-data instructions, schema checks, citation checks; models cannot change the network BUILT
Regulatory compliance	Outage reporting, change records, AI accountability	FCC Part 4 check and ledger BUILT; other jurisdictions PROPOSED
Reliability	The tool fails during the incident it should help with	Model fallback chain, offline rules engine, checkpoints and resume BUILT; HA deployment PROPOSED
Vendor dependence	Model API changes or price rises	Three-tier model fallback and an offline engine BUILT; self-hosted open model option PROPOSED

PART A · SECTION A11

Future development and innovation roadmap

Each item builds on something that exists today. The left column says what is there now, so the proposal is easy to judge.

Area	Built today	Proposed next
Machine learning	Unsupervised detectors; supervised reference reaches F1 0.97 on TelecomTS	Train supervised detectors and root-cause classifiers on incidents engineers confirm in the ledger
Real-time analytics	15-minute KPI windows, live console	Streaming ingestion; per-UE or per-call traces where the operator provides them
Predictive support	Detection after two abnormal windows	KPI forecasting to act before thresholds are crossed (promised in the original pitch, not built)
Automated workflows	Every change needs approval	Pre-approved low-risk change classes with tighter rules, once pilot data supports it
Personalisation	Updates per customer place (home, work, gym)	Channel and language preferences; proactive messages to affected customers only
CRM and ticketing	Internal tickets and credit rule	Connectors to common CRM and ITSM systems
Cloud deployment	Runs locally; Maveric in Docker	Container images, managed database, per-tenant deployment
Enterprise dashboards	Ops console, safety ledger, model card	Management reporting on MTTR, change failure rate and outage minutes
Multilingual support	English	Customer app and voice agent in local languages for emerging markets
Emerging technologies	LTE bands B66, B2, B71; Open M-Plane	5G NR cells, O-RAN rApps on a Non-RT RIC, energy-saving actions, and AI-RAN as it matures

What stays fixed. However much is automated later, the rule that a model can suggest but only code and people can change the network stays. It is the property that makes the rest acceptable to an operator.

PART A · SECTION A12

References

All links accessed on 31 October 2026. Where a figure comes from a secondary report of a paid study, the secondary source is cited and named. Project results come from the repository (README, docs and scripts) at the edition date.

1. GSMA Intelligence. *The Mobile Economy 2025*, infographic, 2025. Worldwide; USD. gsma.com/.../ME_2025_Infographic.pdf
2. Grand View Research, reported by IEEE ComSoc Technology Blog, "Grandview Research: global telecom services market to compound at 6.2% from 2023 to 2030", 23 Nov 2023. Worldwide; USD; fixed and mobile services. techblog.comsoc.org/2023/11/23/...
3. Ericsson Mobility Report, June 2026, reported by TelecomLead, "Global mobile data traffic to reach 515 EB by 2031". telecomlead.com/5g/...-126290
4. Dell'Oro Group. "$160 B in RAN Before 6G", press release, 23 Jul 2025. Worldwide RAN revenue excluding services. delloro.com/news/160-b-in-ran-before-6g
5. McKinsey & Company. "Issue Brief: AI-driven telecom networks", 27 Feb 2026. mckinsey.com/.../issue-brief-ai-driven-telecom-networks
6. TM Forum, "Assessing CSPs' progress toward Level 4 autonomous networks", March 2026, and Accenture study, both reported by Fierce Network, "Telcos pick up the pace to Level 4 autonomous networks". fierce-network.com/cloud/...
7. Federal Communications Commission report on the 22 Feb 2024 AT&T Mobility outage, reported by Fierce Network, "FCC says this is what caused AT&T's big outage in February". fierce-network.com/wireless/fcc-reports-atts-nationwide-outage-february
8. Information Age (ACS). "Optus reveals cause of nationwide outage", Nov 2023. ia.acs.org.au/article/2023/optus-reveals-cause-of-nationwide-outage.html
9. Heavy Reading, commissioned by Spirent, survey of 54 mobile operators (Jan 2016), reported by Voice&Data, "Network outages cost operators more than $20 billion annually", 22 Mar 2016. voicendata.com/...-spirent-report
10. R. Entner (Recon Analytics), Light Reading, "Each carrier nailed its Q2 plan but buried the number that shows the cost", 31 Jul 2026. US; monthly postpaid phone churn. lightreading.com/wireless/...
11. Opensignal. "Retention starts with reception: how mobile network experience drives churn in Australia", April 2025. insights.opensignal.com/2025/04/28/...
12. American Customer Satisfaction Index, Telecommunications Study 2026, reported by ISE Magazine, 19 May 2026. US; 0–100 scale. isemag.com/.../55378638/...
13. IMARC Group. "Self-Organizing Network Market" report page, accessed 31 Oct 2026. Worldwide; USD; 2025 base, 2026–2034 forecast. imarcgroup.com/self-organizing-network-market
14. Global Industry Analysts. "Artificial Intelligence (AI) in Telecommunication", listing by GII, 2024. Worldwide; USD. giikorea.co.kr/report/go1479650-...
15. Research and Markets, reported by IT Brief UK, "AI set to boost telecom market to $11.29bn by 2030", 2 Jan 2024. itbrief.co.uk/story/...
16. Global Market Insights. "AI in Telecommunication Market", Feb 2025. Worldwide; USD. The summary gives two different 2024 values. gminsights.com/industry-analysis/ai-in-telecommunication-market
17. Ericsson. "Ericsson launches Agentic rApp as a Service on AWS", press release, Feb 2026. ericsson.com/en/press-releases/2026/2/...
18. Deutsche Telekom. "Agentic AI for autonomous networks" (RAN Guardian), Feb 2025. telekom.com/en/newsroom/...
19. Nokia. "Nokia introduces MantaRay SON to NTT DOCOMO's multi-vendor 5G network", 25 Nov 2025. globenewswire.com/news-release/2025/11/25/3194000/...
20. 47 CFR 4.9 — Outage reporting requirements, threshold criteria (wireless, paragraph (e)). customsmobile.com/regulations/title47_..._section4.9
21. GSMA. Newsroom boilerplate "uniting more than 750 operators", 16 Nov 2020. gsma.com/newsroom/?p=17741
22. CTIA 2025 Annual Survey highlights, reported by Benton Institute (579 million US wireless connections in 2024). benton.org/headlines/ctia-2025-annual-survey-highlights
23. Fierce Network. "U.S. counts more than 417K cell sites as of 2020". fierce-network.com/wireless/...
24. Feng et al. TelecomTS dataset (MIT licence) and paper, arXiv:2510.06063. github.com/Ali-maatouk/TelecomTS · arxiv.org/abs/2510.06063
25. Competitive Carriers Association. "About CCA". ccamobile.org/about-cca

Project evidence

Detection benchmarks: `scripts/benchmark_simulator.py`, `scripts/benchmark_telecomts.py`. Investigation evaluation: `scripts/eval_investigation.py --live` (31 Oct 2026). Maveric accuracy: `docs/MAVERIC_TWIN.md`. Security tests: `docs/SECURITY_AND_ACCESS_CONTROL.md` section 8. Automated tests: 135 passing at this edition.

RAN Sentinel: technical documentation

Oct 28, 2026 · @QuanTell

1. Overview

RAN Sentinel is an AI co-engineer for the operations centre of a mobile network. It watches a cluster of LTE cells and spots one that is degrading. Meta's Muse Spark agents investigate the cell with read-only tools and propose a fix. Fixed safety rules and a digital twin check that fix before any engineer sees it. The recommendation then goes to the one team that owns it. When a configuration change is approved, it goes out as an Open M-Plane change package with automatic rollback, and the cell is watched until it recovers. Every step is written to the **Evidence-to-Action Safety Ledger**.

It was built by AKMaasif (QuanTell), an LTE RAN engineer, for the Meta Global AI Developer Hackathon (Indie track, theme "Build for Your World"). The code is MIT-licensed.

The problem

US regional and rural carriers run the same kind of LTE network as the national carriers, with a much smaller operations team. For each degraded cell an engineer opens KPI dashboards, alarms and the radio configuration, works out the cause, and decides who should act. A wrong power or tilt change can harm neighbouring cells. Outages also carry federal reporting duties under FCC Part 4 (47 CFR 4.9).

The US setting

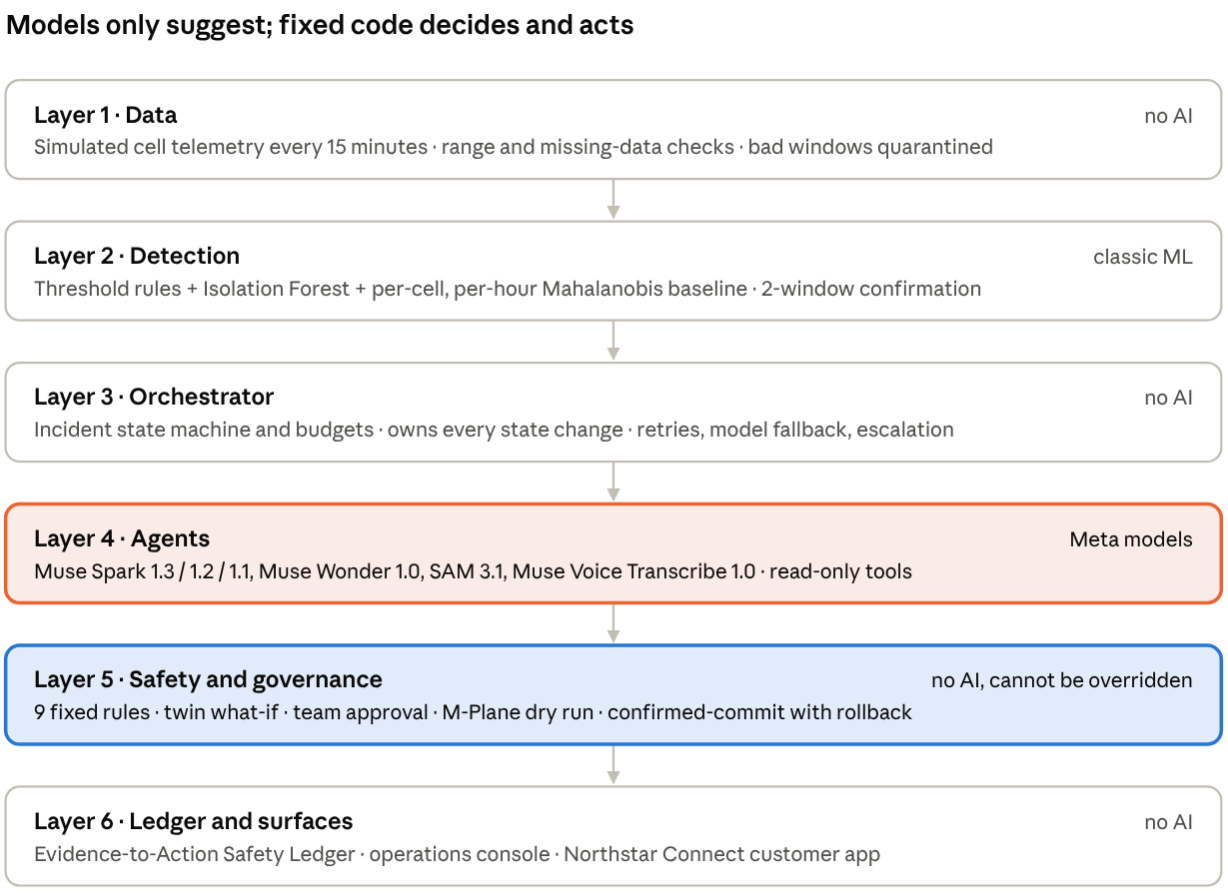
- **Network:** 36 simulated LTE cells (12 sites, 3 sectors each) in Seattle, WA, on US bands B66, B2 and B71.
- **Places:** real Seattle neighbourhoods and ZIP codes, such as Downtown 98101 and Queen Anne 98119.
- **Time:** Pacific time throughout.
- **Regulation:** an FCC Part 4 outage check (section 8).
- **Operator:** the fictional carrier **Northstar Connect**, whose customers see a plain-language status app.

What it is not

Everything runs on a simulated network; nothing connects to a real operator. The FCC check is decision support, not legal advice. No endorsement by Meta, any operator or any equipment vendor is implied.

2. Architecture

RAN Sentinel has six layers. Only layer 4 uses Meta's models, and only layer 3 changes an incident's state. A model can suggest a cause or a plan; it can never apply a change, approve one or skip a rule.



architecture · six layers, AI only in layer 4

Components

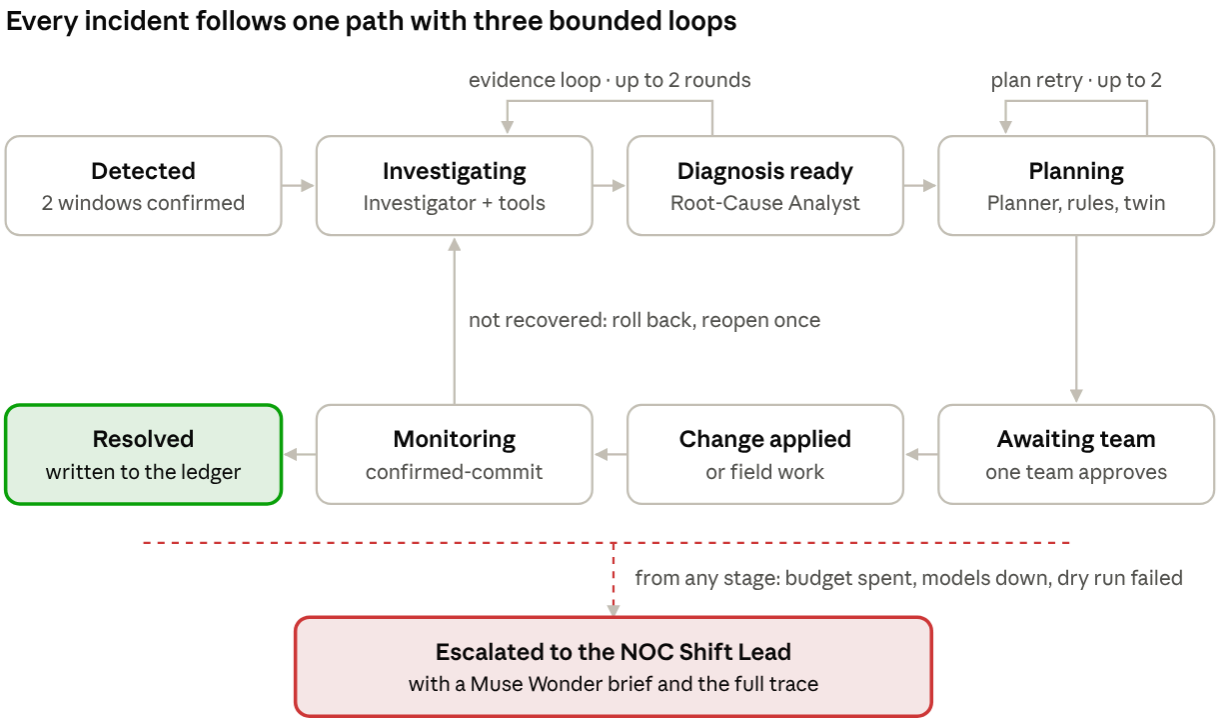
Component	File	Job
Network simulator	`backend/app/simulator.py`	36 Seattle cells, KPIs every 15 minutes, five injectable fault types, alarms

Component	File	Job
Detection	`backend/app/detection.py`	Data checks, three detectors, two-window confirmation
Orchestrator	`backend/app/orchestrator.py`	State machine, budgets, loops, routing, tickets, photos, handover
Agents	`backend/app/agents.py`	Prompts, JSON schemas, citation checks and the offline rules engine
Model provider	`backend/app/providers.py`	Meta Model API client: chat, tools, vision, SAM 3.1, voice; retry and fallback chain
Read-only tools	`backend/app/tools.py`	Anomaly result, KPI window, alarm history, cell configuration, diagnostic checks
Safety validator	`backend/app/safety.py`	Allow-list, nine rules, department routing
Twin	`backend/app/twin.py`	What-if simulation of a change; Maveric adapter
Open M-Plane	`backend/app/mplane.py`	NETCONF change packages, dry run, confirmed-commit and rollback on a simulated radio unit
FCC check	`backend/app/regulatory.py`	Part 4 outage user-minutes and deadlines
Store and ledger	`backend/app/store.py`	SQLite: tickets, ledger, customer updates
API	`backend/app/main.py`	FastAPI endpoints for both apps
Ops console	`frontend/ops/`	Engineer console (vanilla JavaScript)
Customer app	`frontend/connect/`	Northstar Connect status page and assistant

The stack is Python 3.10+ with FastAPI, Uvicorn, NumPy, scikit-learn, httpx and SQLite. The OpenAI Python SDK is the client for Meta's OpenAI-compatible endpoint at `https://api.meta.ai/v1`. The front end is plain HTML, CSS and JavaScript with the IBM Plex fonts, with no build step.

3. End-to-end workflow

The orchestrator owns the incident state machine. The model can suggest; only this code decides a transition.



incident lifecycle · 8 states, 3 loops, 1 side exit

Step by step

1. **Data.** Every 15 minutes the simulator emits KPIs for all 36 cells. Values out of range or missing are quarantined and logged as a data-quality issue; they never reach the detector.
2. **Detection.** Three detectors score each cell (section 6). An alert needs two consecutive windows. A cell with an open incident attaches the new alert to it; neighbouring cells hit at the same time are grouped as related.
3. **Investigation.** The **Investigator** (Muse Spark 1.3) calls five read-only tools: `get_anomaly_result`, `get_kpi_window`, `get_alarm_history`, `get_cell_configuration` (an M-Plane get-config) and `run_diagnostic_check`. A failed tool call is marked as missing evidence, never invented.
4. **Diagnosis.** The **Root-Cause Analyst** ranks four causes (capacity congestion, overshooting coverage, external interference, hardware fault). Every claim must cite an evidence id. If evidence is thin, it asks for more read-only checks from a fixed list: neighbour load, timing-advance distribution, uplink interference scan, VSWR and power readback, config against plan, and handover statistics.

5. **Planning.** The **Remediation Planner** proposes one action (power change, tilt change, or no config change) with prerequisite checks and a rollback. The nine safety rules and the twin check it (section 4), and the **Plan Reviewer** on Muse Spark 1.2 gives an independent second opinion. A failed plan goes back to the planner with the reasons.
6. **Routing.** The plan goes to exactly one team: RAN Optimisation (power or tilt), RF Engineering (interference), Field Operations (hardware), NOC Data Platform (data), NOC Shift Lead (escalations) or Regulatory Compliance (FCC). Nothing changes until that team approves, rejects or asks for more information.
7. **Action.** An approved config change is sent as an Open M-Plane NETCONF package with a dry run first and a confirmed-commit. Field work opens a dispatch ticket instead; the technician can add a site photo.
8. **Monitoring.** The cell is watched. If it recovers, the commit is confirmed and the incident resolves. If not, the radio rolls back by itself and the investigation reopens once.
9. **Records and comms.** Every step goes to the Safety Ledger. The **Customer Comms** agent writes plain-language updates, which a filter checks for internal details before publishing. The **Incident Reporter** writes the report, and **Muse Wonder** writes the shift handover.

Data-quality incidents take a short path: the system re-polls the collector by itself (`AUTO_REMEDIATION`), and only a lasting dropout goes to the NOC Data Platform team.

Budgets and the four loops

Loop	What repeats	Limit
Evidence	Analyst asks for more read-only checks	2 rounds, 8 tool calls per incident
Output repair	Agent JSON fails its schema or cites unknown evidence	1 repair try
Plan revision	Validator or reviewer reasons go back to the planner	2 retries
Recovery	Cell does not recover after a change; rollback and reopen	1 reopen

Error paths

What goes wrong	What happens
Bad or missing telemetry	Window quarantined; a lasting dropout becomes a data-quality incident
Duplicate or related alert	Attached to the open incident, or marked related
Meta API error or timeout	Retry with backoff, then the next model in the chain (1.3, 1.2, 1.1); if all fail, escalate
Invalid model output	One repair try, then escalate
Budget spent	Escalate with the full trace and a Muse Wonder brief
Unsafe plan	Rejected with reasons; replan up to twice, then escalate
Twin unavailable	Plan flagged "not simulated" and needs engineer review
M-Plane dry run fails	Escalate; nothing is committed
Commit not confirmed	The radio reverts by itself
Customer text leaks internal details	Blocked by the filter and replaced by a safe template
Prompt injection inside data	Tool data is passed as data, and the schema and citation checks reject output that follows it

4. Safety and the Safety Ledger

Safety is layer 5. It is plain code with no model, and nothing an agent writes can switch it off. A plan reaches an engineer only after all of these checks.

The nine rules (`backend/app/safety.py`)

#	Rule	What it checks
1	Action in allow-list	Only `adjust_tx_power` , `adjust_tilt` or `no_config_change`
2	Targets the incident cell	A config change must be on the cell under investigation

#	Rule	What it checks
3	Fits the diagnosis	Power or tilt only for capacity or overshoot; interference and hardware get no config change
4	Step within limit	Power up to 3 dB per step, tilt up to 2 degrees
5	Result within bounds	Power 37 to 46 dBm, tilt 0 to 12 degrees
6	Rollback restores current value	The rollback value equals the running value, read over M-Plane, and has a trigger
7	Read-only checks first	At least one prerequisite check from the read-only list
8	Evidence cited	Every cited evidence id exists in this incident
9	Sandbox impact	The twin predicts coverage loss of 1% or less and neighbour PRB use of 85% or less

The other gates

- **Digital twin what-if.** The change is simulated on a copy of the network first. If the twin is down, rule 9 shows "not run" and the plan is flagged for engineer review.
- **Plan Reviewer.** Muse Spark 1.2 reads the plan as an independent second opinion. Its concerns go to the engineer; it cannot approve.
- **Team approval.** Exactly one team receives the plan as a ticket and can approve, reject or ask for more information.
- **M-Plane dry run.** The NETCONF package is validated on the candidate datastore before commit.
- **Confirmed-commit.** The commit carries a confirm timeout of 90 simulated minutes. Monitoring confirms it only if the cell recovers; otherwise the radio rolls back by itself.
- **Red-team switch.** In the demo, a switch forces an aggressive first plan so judges can watch the rules block it and the planner revise it.

The Evidence-to-Action Safety Ledger

Every step is appended to a SQLite ledger: sequence number, incident, wall-clock and simulated time, actor, event, summary and the full payload. Actors include the detector, investigator, analyst, validator, twin, reviewer, orchestrator, monitor, reporter, briefer, compliance and the engineer. Each tool call, model call, rule result, decision, M-Plane message and customer update can be traced from evidence to action. The ledger page in the console filters it by incident and shows the Muse Wonder shift handover.

5. Meta models

Each model has one job where it helps. All of them are called through the Meta Model API at `https://api.meta.ai/v1` with the key from `MODEL_API_KEY` in `.env`.

Model	Agents and jobs	Why this model	If it is unavailable
muse-spark-1.3	Investigator (tool calls), Root-Cause Analyst, Remediation Planner, Customer Comms, Customer Assistant, Incident Reporter, photo reading	Tool calls and JSON output passed on the key	Falls back to 1.2, then 1.1
muse-spark-1.2	Plan Reviewer; fallback for every agent	A different model gives a more independent second opinion	Falls back to 1.3, then 1.1
muse-spark-1.1	Last-resort tier	One more chance before an incident escalates to a person	Escalate to the NOC Shift Lead
muse-wonder-1.0	Escalation Briefer; Shift Handover	The hardest cases get the strongest model; the handover reads every incident and the whole ledger, which needs the 1M-token context	Muse Spark 1.3, then a template
sam-3.1	Field Photo Analyst: boxes antennas, cables and connectors in a technician's site photo	Segmentation is SAM's job; Spark then reads the photo with those boxes	Photo is stored without analysis
muse-voice-transcribe-1.0	Dictated decision notes (ops) and spoken questions (customers)	Engineers on a mast or in a van have their hands full; cell IDs and site names go as keyword hints	Type instead

How the calls work

- **Chat and tools** use the OpenAI-compatible chat completions endpoint. Agent output must be JSON that passes its schema and cites only real evidence ids, with one repair try.

- **Fallback chain.** Each call retries with backoff, then moves down the chain. A three-tier chain (1.3, 1.2, 1.1) runs before escalation.
- **SAM 3.1** uses `POST /v1/responses` with a text prompt and the image as a data URL. It returns boxes as tokens, which the app parses into coordinates with confidence. It tries the request with confidence metadata first and remembers which request shape worked.
- **Voice** uses `POST /v1/asr/transcribe`, a multipart request with a JSON part (model, push-to-talk mode, WAV, keywords) and a mono 16-bit WAV recorded in the browser.

Advisory by design

Voice and photos never change an incident. A dictated note can highlight a decision button (approve, reject, more info, resolved, filed), but the engineer still clicks it. A photo adds evidence and can raise a "does not match the diagnosis" warning.

Deliberately not used

- **muse-image-1.0.** A generated picture is not evidence and could be mistaken for a real site photo.
- **The -contributor variants.** They cost less, but Meta may train on what is sent, and incidents and customer questions can hold sensitive data. An operator can opt in for the customer assistant only with `ASSISTANT_MODEL=muse-spark-1.3-contributor`.

Offline mode

Without a key, the agents use an offline rules engine so the app still runs. The console then says "Offline rules engine" everywhere, so those results cannot be mistaken for Meta-powered ones.

`scripts/check_meta.py` tests every model through the same code the app uses: chat, tool calls, JSON, vision, SAM with a generated image, and transcription with a generated WAV. It prints only the last four characters of the key. Add `--image site.jpg --audio note.wav` to try your own photo and recording.

6. Detection and benchmarks

Detection uses no language model. It reads eight KPIs per cell: PRB utilisation, downlink throughput, CQI, SINR, uplink RSSI, drop rate, handover success and the share of users far from the site. Any of three detectors can raise an alert, and an alert fires only after two consecutive 15-minute windows.

1. **Threshold rules:** seven limits an engineer would set. PRB use above 90%, drop rate above 2%, handover success below 97%, SINR below 6 dB, uplink RSSI above -96 dBm, far-user share above 18%, availability below 99.9%.
2. **Isolation Forest:** learns normal traffic across the cluster.
3. **Cell baseline:** compares each KPI with what is normal for that cell at that hour, scored with a Ledoit-Wolf Mahalanobis distance.

None of them needs labelled faults.

Open dataset: TelecomTS

TelecomTS (MIT licence, ICML 2026) has 32,000 labelled 5G windows with 11 anomaly types. The split was 80/20 train and test. The unsupervised detector was chosen on a validation split taken from the training data, never on test, and thresholds were set for a 1% false-alarm rate. The benchmark ran on your PC because Hugging Face is blocked from the cloud environment.

Detector (TelecomTS test set)	Labels needed	F1	Recall	ROC-AUC	False alarms
Isolation Forest	No	0.23	0.16	0.76	0.9%
PCA reconstruction, context baseline	No	0.52	0.47	0.91	1.3%
Mahalanobis, context baseline (selected)	No	**0.61**	**0.59**	**0.92**	1.4%
Gradient-boosted trees (reference only)	Yes	0.97	0.94	1.00	0.0%

With labels, root-cause classification scored 99.0% accuracy across the 10 synthetic types, and 98.9% after mapping to RAN Sentinel's four causes. These supervised scores are likely optimistic, because TelecomTS anomalies were recorded in separate sessions from normal traffic. Jamming is the hardest type to catch without labels: only 15% was detected. Because labels help so much, the roadmap turns incidents that engineers confirm in the ledger into training labels.

Simulator benchmark

48 injected faults across the five fault types, plus 108 normal cell-days.

Detector	Faults detected	False alarms in 108 normal cell-days
Threshold rules only	79.2%	0
Isolation Forest only	56.2%	0
Cell baseline only	64.6%	0
All three (shipped)	**100% (48 of 48)**	**0**

After the full investigation, the correct root cause ranked first in 97.9% of simulator runs.

Reproduce

```
pip install huggingface_hub
py scripts\benchmark_telecomts.py --export-features
data\telecomts_features.csv.gz
py scripts\benchmark_telecomts.py --features data\telecomts_features.csv.gz
py scripts\benchmark_simulator.py
```

The first command downloads about 1.1 GB once. Later runs use the compact feature file.

7. Open M-Plane and Maveric

Both projects are Meta contributions to LF Connectivity. RAN Sentinel follows their designs, but neither is bundled, and the demo makes clear what is simulated.

	Open M-Plane (`lf-connectivity/open-mplane`)	Maveric (`lf-connectivity/maveric`)
What it is	O-RAN M-Plane NETCONF/YANG stack, in C, MIT licence	AI/ML RAN optimisation testbed with an RF digital twin, in Python, MIT licence
Role in RAN Sentinel	The vendor-neutral way to read radio config and package an approved change	The what-if sandbox for every plan (safety rule 9)

	Open M-Plane (`lf-connectivity/open-mplane`)	Maveric (`lf-connectivity/maveric`)
What is real	The message sequence: `<get-config>` , `<edit-config>` to the candidate datastore, `<validate>` , then `<commit>` with `<confirmed/>` and a confirm timeout. If nobody confirms, the radio reverts. The XML package is shown in the incident.	An adapter that checks whether a Maveric service at `MAVERIC_URL` is reachable
What is simulated	The radio unit is in-process. The XML uses simplified O-RAN-style element names, not the full o-ran-* YANG paths. Electrical tilt is changed directly; real sites often use RET/AISG for tilt.	The prediction itself runs on a copy of the simulator ("fallback twin"), and the result says so. Wiring Maveric's train and simulate calls needs a trained twin of the demo area.

The twin predicts the cell's coverage loss and the offload neighbour's PRB use before and after the change. The limits are coverage loss of 1% or less and neighbour PRB use of 85% or less.

8. FCC Part 4 outage check

`backend/app/regulatory.py` gives a US network operations centre decision support for 47 CFR 4.9(e), wireless outage reporting through NORS.

- **What counts.** Only incidents where the cell loses service (a hardware fault, or availability below 99.9% on a network incident). Slower service is not an outage under Part 4.
- **User-minutes.** Users potentially affected times minutes since discovery. One sector counts as a third of a macro site's average users, set with `FCC_USERS_PER_SITE` in `.env` (default 1,500).
- **Threshold.** At least 30 minutes and at least 900,000 user-minutes makes it reportable.
- **Action.** A reportable outage opens a Regulatory Compliance ticket with three deadlines from discovery: notification within 120 minutes, initial report within 72 hours, final report within 30 days. The incident's FCC panel shows the running count and the deadlines, and a dictated note such as "NORS notification submitted" can mark it filed.

This is decision support for the NOC, not legal advice. With the default of 1,500 users per site, a single sector does not reach the threshold in a short demo; raise `FCC_USERS_PER_SITE` to show a reportable case.

9. Screens

Operations console (`/ops/`)

The console follows the layout of professional network operations (OSS) tools such as Nokia NetAct or Ericsson ENM, without copying any vendor's branding:

- a **top bar** with global search (incident ID, cell ID, site name or ZIP code) and live alarm counters by severity (critical, major, minor, warning);
- a dark **module rail** grouped as Monitor, Operate and Assure;
- one scrolling **work area**;
- a **Demo controls** drawer that opens beside any page;
- a **status bar** with the telemetry window, AI calls and average latency, the twin and the M-Plane link.

Module	What it shows
Monitor · Network overview	KPI tiles with trend sparklines, the cluster map with sectors coloured by health, the incident work queue, network health chart, AI activity board and a live model-call feed, top cells by anomaly score
Monitor · Alarm monitor	Every alarm with X.733-style fields: severity, event type, probable cause, managed object, site, band, ZIP, linked incident. Filter by severity or text, and acknowledge (the ack records who and when)
Monitor · Cell explorer	A network tree by site and band. For each cell, tabs for KPI charts against their limits, alarms, M-Plane radio config against plan, and detector scores
Operate · Incidents	The incident list, and the incident workspace: a fact bar, the workflow timeline, and tabs for diagnosis, plans and safety checks, evidence, field photos, an agent console (a terminal window of every model and tool call), report and ledger. A side column holds the decision panel, detection, FCC and radio config
Operate · Approvals	One card per team with its queue, and the tickets waiting on each
Assure · Safety ledger	The full ledger with an incident filter, and the Muse Wonder shift handover

Module	What it shows
Assure · Model card	Each model's role and fallback, the models not used and why, and the measured accuracy

The backend adds `GET /api/alarms` (the last 400 alarms with event type and probable cause) and `POST /api/alarms/{id}/ack` for the alarm monitor. The layout was checked at desktop (1440 px), tablet (1024 px) and phone (390 px) widths with no horizontal scrolling.

Northstar Connect customer app (`/connect/`)

- **Service status** for Seattle in Pacific time: a ZIP code check, a banner when there is a problem in the customer's area, active updates and the status of every neighbourhood.
- **Ask us:** the Customer Assistant answers questions such as "Is there an issue in 98101?" from published updates only. Customers can type or speak the question.
- Every customer text passes a filter that blocks cell IDs, dBm, tilt, PRB, VSWR, M-Plane, NETCONF, alarm codes and hypotheses. If the model's text fails, a safe template is used.

10. Demo fault controls

Open **Demo controls** from the module rail. Pick a cell (or click a sector on the map), a problem and a severity, then click **Create problem**. Within about two telemetry windows the detector opens an incident and the agents start on their own.

Problem	What happens in the simulator	Expected fix	Team
Capacity surge	An event crowd arrives; PRB use and drops rise	Lower transmit power so users move to a neighbour	RAN Optimisation
Overshoot	A bad config push set the tilt too low; far-user share rises	Restore the tilt	RAN Optimisation
External interference	Uplink RSSI rises, SINR falls	No config fix exists; find the source	RF Engineering
Hardware fault (VSWR)	VSWR and transmit-power alarms, availability falls	Site visit; customers are told a repair is underway	Field Operations

Problem	What happens in the simulator	Expected fix	Team
KPI data dropout	PM files go missing	Re-poll the collector automatically; severity 1 fixes itself	NOC Data Platform

After the incident opens, approve, reject or ask for more information. For field tickets, click **Mark fixed** after dispatch.

Optional switches show the error paths:

- **Red-team the first plan:** forces an aggressive first draft so the safety rules can be seen blocking it.
- **Twin outage:** shows the "not simulated" path.
- **Meta API: primary model down / all models down:** shows the model fallback and the escalation path.

The timed 3-minute video script is in `DEMO.md`.

11. Run, configure and test

Quick start on Windows: double-click `start.bat` in `D:\Meta Ai Heckatho\ran-sentinel`. On the first run it creates the virtual environment, installs the requirements and copies `.env.example` to `.env`. Then it starts the server and opens the console in the browser.

Step by step (Python 3.10 or newer):

```
cd "D:\Meta Ai Heckatho\ran-sentinel"
py -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
copy .env.example .env
notepad .env
py scripts\check_meta.py
py run.py
```

Then open the operations console at http://127.0.0.1:8000/ops/ and the customer app at http://127.0.0.1:8000/connect/. The demo logins are in `.env.example`.

Configuration in `.env` (never committed; listed in `.gitignore`):

Setting	Meaning
`MODEL_API_KEY`	Your Meta Model API key. Leave empty for the offline rules engine
`REVIEWER_MODEL`, `ASSISTANT_MODEL`	Optional model overrides; the file also has PRIMARY_MODEL, FALLBACK_MODEL, LAST_RESORT_MODEL, WONDER_MODEL, SAM_MODEL, VISION_MODEL and VOICE_MODEL, plus token and timeout limits
`MAVERIC_URL`	Optional Maveric service address for the twin adapter
`FCC_USERS_PER_SITE`	Average users per macro site for the FCC estimate (default 1,500)
`OPS_EMAIL`, `OPS_PASSWORD`, `CUSTOMER_EMAIL`, `CUSTOMER_PASSWORD`	Demo logins

Tests: `python -m pytest -q` runs 63 tests in about three minutes. They cover:

- the nine safety rules and M-Plane rollback;
- detection, with no false alerts on a normal day and every fault caught;
- all five fault scenarios end to end;
- the Muse Spark path against a scripted stand-in: tool calls, the repair loop, the evidence loop, the customer-text filter, model fallback and full outage;
- the FCC check, the ZIP-aware assistant, the reviewer and reporter agents and the TelecomTS pipeline;
- the SAM and voice request formats against a mock HTTP server, photo evidence and voice-intent hints;
- the Muse Wonder brief and handover, and the three-tier fallback;
- the alarm monitor fields and acknowledge.

12. Verified and not yet verified

Item	Status	Evidence
Muse Spark 1.3 and 1.2: chat, tool calls, JSON	Verified on a real key	`check_meta.py` on your PC, Oct 28

Item	Status	Evidence
Muse Spark 1.1, Muse Wonder 1.0, Spark vision	Verified on a real key	`check_meta.py`, Oct 28; vision described the test image correctly
SAM 3.1	Verified on a real key	Returned one box with confidence for the generated test image
Muse Voice Transcribe 1.0	Request accepted; accuracy not measured	Tested with a generated tone only, which gives an empty transcript. Next: `check_meta.py --audio note.wav` with a spoken note
SAM on a real site photo	Not yet	Next: `check_meta.py --image site.jpg`
Detection method	Verified offline	TelecomTS benchmark on your PC; simulator benchmark 48 of 48, zero false alarms
End-to-end scenarios, safety, rollback	Verified in tests	63 automated tests pass
Console layout	Verified	Screenshots at 1440, 1024 and 390 px with no JavaScript errors
A full live demo run with the key	Not yet	Do one practice run before recording

Honest limits

- **Simulated network.** 36 cells in Seattle with a small, explainable KPI model. Northstar Connect is fictional.
- **TelecomTS against live data.** TelecomTS checks the detection method on real 5G testbed telemetry (per-UE KPIs every 100 ms). The live demo runs on simulated cell-level KPIs every 15 minutes, because no open dataset offers a cell cluster with config control.
- **Twin.** The what-if runs on a copy of the simulator. The Maveric adapter only checks that the service is running.
- **Open M-Plane.** Simplified O-RAN-style XML goes to an in-process simulated radio unit, not to real hardware or the open-mplane C stack.

- **Support levels.** Hypothesis support levels describe evidence strength, not calibrated probabilities.
- **FCC.** Decision support only, not legal advice.
- **No endorsement.** No Meta, operator or vendor endorsement is implied.

13. Timeline

Date	What was done
Oct 26	Hackathon build window opens.
Oct 27	Reviewed the 22-section design doc and the planning chat. Confirmed the Meta key worked with muse-spark-1.3. Wrote the demo storyboard and the layered workflow v2 (6 layers, 4 bounded loops, error paths). Researched Meta's telecom work and planned the Maveric and Open M-Plane integration. Built the first working app from your UI mockup: simulator, detection, orchestrator, agents, safety, M-Plane, ledger, ops console and customer app.
Oct 28, early	Moved the network to the US: Seattle sites and ZIP codes, bands B66, B2 and B71, Pacific time, the FCC Part 4 check and a Regulatory Compliance desk. Added the Plan Reviewer, the Incident Reporter, the Model card page and the benchmark scripts. Wrote the business case.
Oct 28, morning	You ran the TelecomTS benchmark on your PC. The plain Isolation Forest was weak, so the per-cell Mahalanobis baseline was added; simulator detection reached 48 of 48 with zero false alarms.
Oct 28, 08:08 to 08:35 UTC	Added every useful Meta model: Spark 1.1 as the last-resort tier, Muse Wonder for briefs and the handover, SAM 3.1 and Spark vision for site photos, Muse Voice Transcribe for dictated notes and spoken questions. You ran `check_meta.py` and every model passed.

Date	What was done
Oct 28, 10:23 to 11:45 UTC	Checked the work against Meta's public hackathon requirements. Redesigned the console in the style of professional OSS tools, with the alarm monitor, cell explorer, tabbed incident workspace, agent console, approvals and status bar. Added LICENSE, SUBMISSION.md, DEMO.md, start.bat and the alarm test (63 tests). Wrote the requirements check.
By Nov 3	Your target: practice run, public GitHub repo, 3-minute video.
Nov 4, 9:00 PM ET	Submission deadline (Nov 5, 7:30 AM in Sri Lanka). Judging runs Nov 4 to 8; finals are on Nov 9.

Related documents: the hackathon requirements check and the business case. Source files: `README.md`, `SUBMISSION.md`, `DEMO.md` and `docs/workflow.mmd`.